# Paper III – Subspeciality of Urology (MCh Urology, Winter 2017)

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## III-1. Describe embryology of penis. Describe management of proximal hypospadias.

**Type:** LAQ (25 marks)

### A. Embryology of the penis

**Indifferent stage (up to 8–9 weeks)**

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

**Hormonal control**

- SRY (sex-determining region Y) → testis → **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

| Primordium (germ layer) | Male | Female |
|---|---|---|
| Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce) | Glans, corpora | Clitoris |
| Urethral folds | Fuse → penile urethra, median raphe | Labia minora |
| Labioscrotal swellings | Scrotum | Labia majora |
| Urethral plate (endoderm) | Whole urethra, including glanular | Vestibule |

**Urethral formation: the "double zipper"**

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe.

- The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated.
- Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks.
- **Prepuce** fuses dorsally first; ventral fusion awaits urethral closure, hence the **dorsal hood** in hypospadias.
- Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site).
- **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

## B. Proximal hypospadias: evaluation

- Meatus proximal shaft, penoscrotal, scrotal or perineal (10–25%). Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone). A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- No routine VCUG (voiding cystourethrogram) or renal ultrasound; prostatic utricle commoner.

## C. Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects transient; evidence mixed, but two RCTs (randomised controlled trials) showed fewer complications.

## D. Curvature correction (decides one- vs two-stage)

```
Deglove + excise dysplastic dartos + artificial erection
   ├─ straight        → urethroplasty
   ├─ residual <30°   → dorsal midline plication (12 o'clock)
   └─ >30°            → divide urethral plate
                          └─ still >30° → ventral corporotomy ± graft → staged repair
```

| | Dorsal plication (Nesbit, Baskin) | Ventral lengthening |
|---|---|---|
| Method | Shortens the convex dorsal side | 1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft |
| Use | Mild (≤15–30°) | Moderate–severe |
| Recurrence | 28% | 9% |

## E. Urethroplasty

| Approach | Technique | Comment |
|---|---|---|
| One-stage **proximal TIP (tubularised incised plate)** | Plate kept, incised and tubularised | Recurrent curvature, stricture: largely abandoned for severe cases |
| One-stage **onlay island flap** | Inner preputial flap on preserved plate | Diverticulum, fistula |
| One-stage **TPIF (transverse preputial island flap, Duckett tube)** | Tube after plate division | High complications |
| **Two-stage Bracka / STAG (staged tubularised autograft)** | Stage 1: correct curvature, divide plate, quilt inner-prepuce or buccal graft. Stage 2 (≥6 months): tubularise, barrier layer | **Preferred for most proximal cases**; STAG lowest complications (about 23%) |
| **Two-stage Byars** | Dorsal preputial flaps moved ventrally, tubularised later | Complications 43–81% |

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; scrotoplasty for transposition.

## F. Complications and outcomes

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** commonest (repair ≥6 months later; exclude distal obstruction), glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, diverticulum (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans).
- A long, rigid neourethra raises resistance (Poiseuille), causing fistula and diverticulum.
- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

**Diagram to draw:** double-zipper urethral formation; curvature algorithm.

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias plus undescended testis → DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka preferred; fistula commonest complication.

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# III-2. Describe anatomy of pelvic floor in females and describe surgeries for SUI in females.

**Type:** LAQ (25 marks)

## Introduction

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, coughing or sneezing. Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.

## A. Anatomy of the female pelvic floor

### 1. Pelvic diaphragm

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops behind anorectal junction | Anorectal angle |
| Pubococcygeus (pubovaginal) | Pubis, ATFP (arcus tendineus fasciae pelvis) → viscera, raphe | Slings urethra and vagina |
| Iliococcygeus | Arcus tendineus levator ani, ischial spine → coccyx | Horizontal **levator plate** |
| Coccygeus (not levator) | Ischial spine → sacrum, coccyx | Lies on sacrospinous ligament |

- Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, fascia then carries the load.

### 2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

### 3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

### 4. DeLancey levels (1992)

| Level | Structures | Defect |
|---|---|---|
| I suspension | Cardinal–uterosacral | Apical prolapse, enterocele |
| II attachment | ATFP, levator fascia | Cystocele (central or paravaginal), rectocele |
| III fusion | Perineal membrane, perineal body | **Urethral hypermobility**, perineal descent |

### 5. Continence mechanism

- Continence zone: proximal and mid-urethra. **Oestrogen-dependent vascular submucosa** gives much of closure pressure; smooth and striated sphincters; pudendal nerve (S2–4) drives the guarding reflex.

| Theory | Mechanism | Surgical answer |
|---|---|---|
| Pressure transmission (Enhorning 1961) | Hypermobile urethra leaves the abdominal pressure zone | Burch |
| Hammock (DeLancey 1994) | Urethra compressed against a firm suburethral layer | Restore backing |

| Theory | Mechanism | Surgical answer |
|---|---|---|
| Integral theory (Petros & Ulmsten 1990) | Pubourethral ligament laxity at mid-urethra | Mid-urethral sling |
| ISD (intrinsic sphincter deficiency) | Urethra fails to close (VLPP (Valsalva leak point pressure) <60, MUCP (maximum urethral closure pressure) <20 $cmH_2O$) | Sling, bulking, AUS (artificial urinary sphincter) |

## B. Before surgery

- Index patient: cough stress test, PVR (post-void residual), urinalysis; urodynamics if SUI not demonstrated or complicated. **PFMT (pelvic floor muscle training)**, weight loss, pessary first.

## C. Surgeries for SUI

| Operation | Principle | Key facts |
|---|---|---|
| **MUS (mid-urethral sling)**: retropubic TVT (tension-free vaginal tape, Ulmsten 1995); transobturator (Delorme 2001; inside-out de Leval) | Loose type I polypropylene tape under the **mid-urethra**; urethra kinks on it during stress | **First line** (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) strong recommendation). **TOMUS**: 80.8% vs 77.7%, equal |
| **SIMS (single-incision mini-sling)** | Vaginal only | Non-inferior to transobturator for subjective cure; conditional recommendation (2023) |
| **Autologous PVS (pubovaginal sling)** | Rectus fascia or fascia lata at the **bladder neck**, minimal tension | All types including **ISD**, failed MUS, mesh-averse. **SISTEr**: sling beat Burch (47% vs 38%) with more voiding dysfunction and UTI (urinary tract infection) |
| **Burch colposuspension** (open, laparoscopic, robotic) | Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's ligament** | 85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD |
| Paravaginal repair, VOS (vagino-obturator shelf) | Vagina to ATFP or obturator fascia | Flatter, less obstructive |
| MMK (Marshall–Marchetti–Krantz) | Bladder neck to symphysis periosteum | Osteitis pubis 0.7–2.5%; abandoned |
| **Bulking agents** | Submucosal injection improves coaptation | Frail, minimal surgery; repeat injections |
| **AUS** | Bladder-neck cuff | Second line for severe recurrent ISD; erosion, revision |

- Needle suspensions abandoned (pull-through); anterior colporrhaphy is not an SUI operation.

**Burch steps:** Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

### Retropubic vs transobturator MUS

| | Retropubic | Transobturator |
|---|---|---|
| Bladder perforation | 4.5% | 0.6% |
| Voiding dysfunction | More (TOMUS surgery 2.7% vs 0%) | Less |
| Groin pain | 1.3% | 6.4% |

| | Retropubic | Transobturator |
|---|---|---|
| ISD, long-term durability | Often preferred | Possibly more repeat surgery |

## D. Complications

- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal) → urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS (International Urogynecological Association/International Continence Society) terms): exposure, extrusion, perforation into bladder or urethra, pain → excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.
- **Also know:** CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension).

**Diagram to draw:** DeLancey levels and hammock cross-section with Burch, bladder-neck and mid-urethral sling sites.

> **Key points to remember:**
> - Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
> - DeLancey: I suspension, II attachment (ATFP), III fusion.
> - MUS first line; TOMUS: retropubic = transobturator.
> - SISTEr: sling > Burch; PVS for ISD; urethral injury → no mesh.

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# III-3a. Sacral Neuromodulation.

**Type:** SAQ (10 marks)

## Definition and mechanism

- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator).
- It acts on **afferents** to the forebrain, not motor nerves. In OAB (overactive bladder) it damps PMC (pontine micturition centre) firing; in retention it restores filling awareness, so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

## Indications (third line, AUA (American Urological Association) and EAU (European Association of Urology))

| FDA (Food and Drug Administration) approved | Not approved |
|---|---|
| Refractory UUI (urgency urinary incontinence) 1997 | Bladder pain syndrome |
| Urgency–frequency, **non-obstructive retention** 1999 | Neurogenic bladder (small studies positive) |
| Faecal incontinence | Pelvic pain |

- **Fowler's syndrome** (young women in retention) responds well.

## Technique

| Test | PNE (percutaneous nerve evaluation) | Staged implant |
|---|---|---|
| Lead | Temporary, office, local anaesthesia | Permanent tined lead, theatre, fluoroscopy |
| Duration | 5–7 days | Up to 4 weeks |
| Use | OAB | Higher conversion; retention |

- Prone; S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

| Root | Pelvic response | Foot response |
|---|---|---|
| S2 | — | Calf and foot rotation |
| **S3** | **Bellows**, perineal sensation | **Great-toe flexion** |
| S4 | Bellows only | None |

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock; 10–16 Hz at sensory threshold.

## Outcomes

- **InSite**: success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years.
- **ROSETTA**: onabotulinumtoxinA 200 U slightly better for UUI (−3.9 vs −3.3 episodes/day) and higher satisfaction, but more UTI (urinary tract infection) and self-catheterisation.

## Complications and contraindications

- Pain at IPG site, lead migration, loss of efficacy, infection; **revision over 30% at 5 years**; costly; battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

## Recent advances

- Rechargeable IPGs (≥15 years); **full-body MRI (magnetic resonance imaging)-conditional** systems; bilateral stimulation no better; PTNS (percutaneous tibial nerve

stimulation) as alternative.

**Diagram to draw:** lead in S3 foramen with IPG in the buttock.

> **Key points to remember:**
> - Afferent modulation treats OAB and retention.
> - S3: bellows plus great-toe flexion.
> - Implant if test improves ≥50%; revision >30% at 5 years.

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# III-3b. Botulinum toxin

**Type:** SAQ (10 marks)

## Introduction

- *Clostridium botulinum* neurotoxin, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**; type B is short-acting.
- 150 kDa: heavy chain (100 kDa) plus zinc-dependent light chain (50 kDa).

## Mechanism

1. Heavy chain binds the cholinergic terminal; toxin is internalised.
2. Light chain **cleaves SNAP-25 (synaptosomal-associated protein 25)**, a SNARE (vesicle-docking) protein, so vesicles cannot fuse and **acetylcholine release stops**.

- **Afferent action**: less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) receptors, so urgency falls.
- Effect wanes as terminals sprout (6–9 months).

## Technique

- Cystoscopic injection into detrusor at **20 sites (100 U) or 30 sites (200 U)**, usually sparing trigone; sterile urine, antibiotic cover; PVR (post-void residual) at 2 weeks.

## Urological uses

| Indication | Dose and evidence |
|---|---|
| **Idiopathic OAB (overactive bladder)**, third line | **100 U** (EMBARK): incontinence −59%, about 25% dry. **ABC**: equal to antimuscarinic, more complete dryness. **ROSETTA**: 200 U ≥ SNM (sacral neuromodulation), more UTI (urinary tract infection) |
| **NDO (neurogenic detrusor overactivity)**: SCI (spinal cord injury), MS (multiple sclerosis) | **200 U** (DIGNITY): lower detrusor pressure, larger capacity |
| **Paediatric NDO** (spina bifida) | About **10 U/kg (maximum 360 U)** in Campbell series; FDA label (2021, age ≥5) is 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation |
| **DSD (detrusor–sphincter dyssynergia)**, underactive detrusor | Intrasphincteric 50–100 U; men unable to do CISC (clean intermittent self-catheterisation) |
| **IC/BPS (interstitial cystitis/bladder pain syndrome)** | 100 U, often trigonal; AUA (American Urological Association) option, off-label |
| **BPH (benign prostatic hyperplasia)** | Intraprostatic: early small trial positive, but larger placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo**; not recommended |

- **Also know:** paediatric dysfunctional voiding (sphincter); pelvic floor spasm; ED (erectile dysfunction) trials.

## Complications

- **Retention needing CISC**: 0–12% in OAB trials; real-world about 20% women, up to 35% men; higher in NDO. Each 10 mL rise in baseline PVR raises risk about 27%. **Patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

- Contraindications: active UTI, myasthenia gravis, pregnancy. Repeat injections stay effective.

**Diagram to draw:** cholinergic synapse showing SNAP-25 cleavage.

> **Key points to remember:**
> - Cleaves SNAP-25; efferent and afferent action.
> - 100 U idiopathic OAB; 200 U NDO; children 10 U/kg (max 360 U) in series, FDA label 6 U/kg (max 200 U).
> - Main risks: retention needing CISC, UTI.

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# III-3c. Tacrolimus.

**Type:** SAQ (10 marks)

## Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

## Mechanism

```
Tacrolimus + FKBP12 (FK506-binding protein 12)
   → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase)
   → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus
   → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)
```

- Ciclosporin binds **cyclophilin** instead; same end result.

## Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

## Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

## Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

## Toxicity

| System | Effect |
|---|---|
| Kidney | **Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy) |
| Metabolic | **NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia |
| Neurological | **Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome) |
| Others | Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder)) |

## Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

## Versus ciclosporin

| | Tacrolimus | Ciclosporin |
|---|---|---|
| Acute rejection | Lower | Higher |
| Diabetes, tremor, alopecia | More | Less |
| Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension | Less | More |

- **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens.
- **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis).

**Diagram to draw:** T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - FKBP12 → calcineurin → NFAT → IL-2 blocked.
> - CYP3A4/P-gp; trough-guided; many interactions.
> - Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
> - ELITE-Symphony: low-dose tacrolimus best.

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# III-3d. Retrocaval ureter.

**Type:** SAQ (10 marks)

## Definition

- The **right ureter** passes **behind the IVC (inferior vena cava)**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.

## Embryology

- Paired fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney.
- Normally the right **supracardinal** vein, **dorsal** to the ureter, forms the infrarenal IVC, so the ureter lies in front.
- If a vein **ventral** to the ureter persists as the IVC (lumbar subcardinal in Campbell; "persistent posterior cardinal vein" in many texts), the ureter is trapped behind it.
- Variant: double right IVC (ureter between limbs).

## Incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis. Associations: horseshoe kidney, left renal anomalies.

## Classification (Bateson & Atkinson)

| | Type I (low loop), commoner | Type II (high loop) |
|---|---|---|
| Crossing | Lower, at L3–L4 | Higher, at renal pelvis level |
| Shape | **"Fishhook" / reverse-J**, S-curve | Smooth horizontal **sickle** curve |
| Obstruction | Moderate–severe, at iliopsoas edge | Mild or none (pelvis also crosses) |

## Investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) alternative.
- IVU (intravenous urography): fishhook, ureter lost beyond the J hook. Retrograde: **S-curve** at L3–L4. **Diuretic renogram**: obstruction, split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**; pyeloplasty fails.

## Management

- **Non-obstructed (often type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it anterior to the IVC, spatulated tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; preserve blood supply.
- **Laparoscopic or robotic** (transperitoneal or retroperitoneal) now preferred: equal success, faster recovery. Nephrectomy if non-functioning.
- **Also know:** retroiliac ureter (L5–S1, vasal anomalies); mimics: spermatic or lumbar vein, psoas tendon.

**Diagram to draw:** right ureter hooking behind the IVC with fishhook pelvis.

**Key points to remember:**

- Venous anomaly; right side; 1 in 1500; male 3:1.
- Type I fishhook (obstructs); type II sickle.
- CT urography; repair by anterior transposition and ureteroureterostomy.

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# III-3e. DVIU (Direct Visual Internal Urethrotomy).

**Type:** SAQ (10 marks)

## Definition and principle

- Incision of stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged.
- Healing is by **secondary intention**: a **race between epithelialisation and wound contraction**. If contraction wins, the stricture recurs. Deep spongiofibrosis cannot be cured endoscopically.

## Indications (favourable)

| Favourable | Unfavourable |
|---|---|
| **Bulbar** | Penile, fossa navicularis, any non-bulbar |
| **Short (<1.5–2 cm)** | Longer |
| Superficial, no dense spongiofibrosis | Dense scar (straddle, LS (lichen sclerosus)) |
| **First procedure** | Repeat; luminal narrowing >74% |

## Technique

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED (erectile dysfunction), fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit.

- **Laser** (holmium, Nd:YAG (neodymium:yttrium–aluminium–garnet), others): **no advantage over cold knife**.

## Outcomes

- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none.
- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Repeat DVIU lengthens scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure.

## Guideline position

- **AUA (American Urological Association) 2016:** DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **urethroplasty, not repeated endoscopy**, after failure; no permanent stents.
- Reconstructive ladder is abandoned.

## Adjuncts after DVIU

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C, steroid injection (limited data); UroLume withdrawn.

## Complications

- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI (urinary tract infection), sepsis; ED; spongio-cavernous fistula; incontinence if sphincter incised.

## Recent advance

- **Optilume** paclitaxel-coated balloon (FDA (Food and Drug Administration) 2023): **ROBUST III** stricture-free 74.6% vs 26.8% at 1 year after failed endoscopy.

**Diagram to draw:** urethrotome cutting at 12 o'clock over guidewire.

> **Key points to remember:**
> - Best for bulbar, <1.5–2 cm, first attempt.
> - Santucci: ~8% long-term success; repeat DVIU futile.
> - AUA: urethroplasty after failure; laser = cold knife.

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## III-3f. TUR Syndrome.

**Type:** SAQ (10 marks)

### Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). About 1–2% in large series; falling.

### Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, water (plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity**: ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Even routine resection absorbs 800–1000 mL.

### Risk factors

- Gland **>45 mL**, resection **>90 minutes** (cooperative study), high irrigation pressure, capsular perforation, sterile water.

### Clinical features

- Awake patient: restlessness, confusion, **nausea, vomiting**, **visual disturbance**.
- Cardiovascular: **hypertension and bradycardia**, later hypotension, tachypnoea, ECG (electrocardiogram) changes.
- Severe: seizures, coma, death. **Na <120 mEq/L** is dangerous.

### Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT (number needed to treat) about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time (stage large glands); check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol tagging** (breath test) monitors absorption.

### Treatment

1. Stop surgery, achieve haemostasis, insert catheter.
2. ABC (airway, breathing, circulation); oxygen; monitor; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes up to 300 mL. **Target rise 4–6 mEq/L.**
5. Correction limit: up to **12 mEq/L in 24 h** for acute hyponatraemia (8–10 mEq/L if symptoms begin >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy; sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

**Diagram to draw:** flowchart: absorption → overload + hyponatraemia → cerebral oedema.

**Key points to remember:**

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

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# Rapid Revision Sheet – Paper III

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**III-1. Describe embryology of penis. Describe management of proximal hypospadias. (LAQ)**

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias plus undescended testis → DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka preferred; fistula commonest complication.

**III-2. Describe anatomy of pelvic floor in females and describe surgeries for SUI in females. (LAQ)**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- MUS first line; TOMUS: retropubic = transobturator.
- SISTEr: sling > Burch; PVS for ISD; urethral injury → no mesh.

**III-3a. Sacral Neuromodulation. (SAQ)**

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves ≥50%; revision >30% at 5 years.

**III-3b. Botulinum toxin (SAQ)**

- Cleaves SNAP-25; efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO; children 10 U/kg (max 360 U) in series, FDA label 6 U/kg (max 200 U).
- Main risks: retention needing CISC, UTI.

**III-3c. Tacrolimus. (SAQ)**

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
- ELITE-Symphony: low-dose tacrolimus best.

**III-3d. Retrocaval ureter. (SAQ)**

- Venous anomaly; right side; 1 in 1500; male 3:1.
- Type I fishhook (obstructs); type II sickle.
- CT urography; repair by anterior transposition and ureteroureterostomy.

**III-3e. DVIU (Direct Visual Internal Urethrotomy). (SAQ)**

- Best for bulbar, <1.5–2 cm, first attempt.
- Santucci: ~8% long-term success; repeat DVIU futile.
- AUA: urethroplasty after failure; laser = cold knife.

**III-3f. TUR Syndrome. (SAQ)**

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.